SYNTHETIC DATA — 교육용 가상 데이터이며 실제 제품 사양·운영 기준과 무관합니다.

MCCB 100AF (MCCB-ABS-103c) — Datasheet

1. Overview

Molded case circuit breaker, 100A frame, thermal-magnetic trip unit. Product family: Low-voltage circuit breakers. Lifecycle status: **ACTIVE**.

2. Technical specifications

2.1. Ratings

Item	Value
Frame size	100AF
Rated current (In)	15, 20, 30, 40, 50, 60, 75, 100 A
Rated operational voltage (Ue)	AC 690 V max
Number of poles	2P / 3P / 4P

2.2. Performance and characteristics

Item	Value
Ultimate breaking capacity Icu at AC 415 V	37 kA
Service breaking capacity Ics	75% of Icu

2.3. Endurance and communication

Item	Value
Mechanical endurance	8,500 operations
Electrical endurance	1,500 operations at In

2.4. Protection and options

Item	Value
Magnetic trip setting	Fixed, 10 x In

3. Installation data

3.1. Terminals

- Terminal tightening torque: 3.5 N·m (M5)
- Dimensions (W x H x D): 75 x 130 x 60 mm
- Weight: 0.9 kg
- Degree of protection: IP20 (front IP40)

- Operating temperature: -5 to +40 °C (derating above 40 °C)

4. Standards and certifications

Applicable standards:

- IEC 60947-2

Certifications held:

- CE
- CB
- KC

Certificates and declarations of conformity are available from the Certification & Compliance team on request. Certifications not listed above are not held for this product; project-specific evaluation can be requested.

5. Ordering information

Item	Value
Ordering code	MCCB-ABS-103c
Standard lead time	4 weeks ex-works
Minimum order quantity	10
Warranty	18 months (see POL-WARRANTY)
Firmware	not applicable

6. Related documents

- IG-MCCB-ABS-103c — Installation and Commissioning Guide
- TG-MCCB-ABS-103c — Troubleshooting and Error Code Guide
- MG-MCCB-ABS-103c — Maintenance Guide
- FAQ-LV_BREAKER — Low-voltage circuit breakers FAQ

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MCCB 100AF (MCCB-ABS-103c) — Installation and Commissioning Guide

1. Safety

Installation, wiring and commissioning must be performed by qualified personnel. Disconnect and lock out all power sources before work. Verify absence of voltage.

2. Receiving and storage

Inspect the packaging on arrival and report damage within 7 days (see POL-RETURN). Store indoors in a dry place between -25 and +55 °C, relative humidity below 85% without condensation.

3. Mounting and clearances

- Required clearances: 50 mm above and below the breaker for arc gases; 10 mm between adjacent breakers
- Operating temperature: -5 to +40 °C (derating above 40 °C)
- Degree of protection: IP20 (front IP40)
- Above 1,000 m altitude, apply derating per the datasheet.

4. Wiring and tightening torques

- Terminal tightening torque: 3.5 N·m (M5)
- Use copper conductors rated for at least 75 °C.
- Separate power and signal cables by at least 200 mm.

5. Commissioning procedure

1. Check the rating plate against the single-line diagram
2. Verify tightening torque of all terminals
3. Measure insulation resistance (≥ 5 MΩ at 500 V DC)
4. Operate the breaker manually ON/OFF three times
5. Check trip unit settings against the coordination study

6. Pre-energisation checklist

☐ Product rating matches the drawing
☐ All terminals tightened to specified torque
☐ Protective earth connected
☐ Insulation resistance measured and recorded
☐ Settings / parameters recorded

7. Common commissioning issues

Symptom	Likely cause	See
Breaker trips at motor start	Motor inrush exceeds magnetic trip setting	TG-MCCB-ABS-103c §3.1
Terminal overheating / discoloration	Loose connection or undersized cable	TG-MCCB-ABS-103c §3.2
Breaker cannot be reset after trip	Handle not moved to reset position before ON	TG-MCCB-ABS-103c §3.3

On-site commissioning support can be requested through the Global CS Center. For low-voltage circuit breakers the request is handled by the technical support team.

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MCCB 100AF (MCCB-ABS-103c) — Maintenance Guide

1. Maintenance intervals

Task	Interval
Visual inspection and cleaning	12 months
Terminal torque check	12 months
Manual ON/OFF operation	6 months
Test button (ELCB)	monthly

2. Spare parts

- No user-replaceable parts; replace the complete unit.

3. End of life

Lifecycle status: ACTIVE. Expected service life is 20 years under normal conditions. Dispose of according to local regulations for electrical equipment.